

SepsisWatch

Deep Learning and Computer Vision for Early Sepsis Prediction
in Intensive Care Units — Technical Report

Project	SepsisWatch (educational / research project)
Domain	Clinical decision support, ICU monitoring
Core techniques	BiGRU + Attention, Gramian Angular Field imaging, CNN, model fusion
Dataset	PhysioNet / Computing in Cardiology Challenge 2019
Report version	1.0 — 2026

Important note on scope: This report documents the system design, methodology, and validation plan for an educational/research AI project. It does not claim clinical validation, regulatory approval, or real-world deployment outcomes. Any performance figures cited from external sources are clearly attributed; no fabricated results are presented as our own.

Abstract

Sepsis is a leading cause of in-hospital mortality, and early detection materially improves outcomes. This report presents SepsisWatch, an educational, end-to-end system for early sepsis risk prediction from ICU time-series data. The system combines two complementary branches: a recurrent temporal model (bidirectional GRU with attention) that captures the trajectory of a patient's vital signs over a 12-hour window, and a computer-vision branch that transforms the same readings into Gramian Angular Field (GAF) and recurrence-plot images, analyzed by a convolutional neural network (CNN). The two branches are fused into a single risk score. The system is built around the PhysioNet/Computing in Cardiology Challenge 2019 dataset (40,336 publicly available ICU patient records). We describe the dataset, methodology, explainability mechanism, deployment architecture, monitoring stack, and — critically — an honest discussion of what clinical validation would require and what has not yet been done. We conclude with a concrete future-work roadmap covering real-time streaming, EHR integration, federated learning, transformer architectures, and multi-hospital validation.

1. Problem Statement

Sepsis is a dysregulated host response to infection that can rapidly progress to organ failure and death. In intensive care settings, clinical recognition of sepsis often lags behind the physiological onset of deterioration by several hours — a delay associated with materially worse outcomes. The Sepsis-3 consensus definition (Singer et al., 2016) formalizes sepsis as a suspected infection accompanied by an acute increase of at least two points in the Sequential Organ Failure Assessment (SOFA) score.

The central technical problem addressed here is: given a rolling window of hourly vital-sign and laboratory measurements for an ICU patient, can a model predict — several hours in advance — whether that patient is trending toward a sepsis diagnosis, while keeping the false-alarm rate low enough to remain clinically usable? This is fundamentally a multivariate, irregularly-sampled, highly imbalanced time-series classification problem (sepsis-positive hours represent roughly 1.8% of all recorded hours in the public dataset).

1.1 Why this is hard

- Severe class imbalance: naive classifiers trivially achieve high accuracy by always predicting 'no sepsis'.
- Missingness: many lab values are measured intermittently, not at every hour.
- Label lag: the Sepsis-3 label itself is a retrospective clinical judgment, not a ground-truth 'moment' of onset.
- Clinical cost asymmetry: a missed sepsis case is far costlier than a false alarm, but too many false alarms cause alert fatigue and erode trust.

2. Literature Review

2.1 The PhysioNet/CinC Challenge 2019

The PhysioNet/Computing in Cardiology Challenge 2019 (Reyna et al., 2019, published in Critical Care Medicine) established the task this project builds on: hourly, six-hour-ahead sepsis prediction from ICU time series, evaluated with a custom clinical utility function that rewards early true detections and penalizes both false alarms and late detections. The challenge released de-identified records from two hospital systems (Beth Israel Deaconess Medical Center and Emory University Hospital) publicly, with a third, held-out hospital system used only for blind evaluation. 104 teams submitted 853 algorithms; 88 were successfully evaluated on the full test set. The winning team ('SailOcean') achieved a final utility score of 0.364, with per-hospital scores of 0.430, 0.422, and -0.048 on systems A, B, and C respectively (Bennett et al., 2021, post-challenge analysis, arXiv:2012.11013) — illustrating that generalization to an unseen hospital remains a hard open problem even for top-performing approaches.

2.2 Recurrent and attention-based models for clinical time series

GRU and LSTM architectures remain competitive baselines for irregularly-sampled clinical time series when combined with careful imputation and masking strategies. Attention mechanisms over the temporal axis provide a natural, cheap form of interpretability by surfacing which time steps most influenced a prediction — the same mechanism used in SepsisWatch's GRU branch.

2.3 Time-series imaging for CNN-based classification

Encoding a univariate time series as a 2D image — via Gramian Angular Fields (Wang and Oates, 2015) or recurrence plots — allows standard 2D CNN architectures (originally developed for natural images) to be applied to time-series classification. GAF images preserve temporal correlation as pairwise angular relationships, while recurrence plots highlight periods of instability or repeated states. This project applies both transforms to the four most clinically salient vitals (heart rate, oxygen saturation, mean arterial pressure, respiration rate) and feeds the resulting multi-channel image to a small ResNet-style CNN, on the hypothesis that visual instability patterns (e.g., increasing HR variability paired with a falling MAP trend) carry signal that is complementary to what a purely sequential model captures.

2.4 Explainable AI in clinical decision support

SHAP (SHapley Additive exPlanations, Lundberg and Lee, 2017) is widely used to attribute a model's output to individual input features using a game-theoretic formulation, and is one of the few explanation methods with axiomatic consistency guarantees. In clinical decision support, feature-level explanations (e.g., 'low MAP' and 'high heart rate' contributed most to this risk score) are considered a baseline requirement for clinician trust and adoption, and are increasingly expected by regulators evaluating software-as-a-medical-device submissions.

3. Dataset

SepsisWatch is built around the PhysioNet/Computing in Cardiology Challenge 2019 dataset, one of the largest publicly available, peer-reviewed ICU time-series datasets for sepsis prediction.

Metric	Value	Source
Public patients (Set A + Set B)	40,336	physionet.org/content/challenge-2019
Hospital — Set A	Beth Israel Deaconess Medical Center	Challenge data description
Hospital — Set B	Emory University Hospital	Challenge data description
Total patients across 3 hospital systems	> 60,000	Reyna et al., 2019 (CCM)
Variables per patient	up to 40 (vitals, labs, demographics)	Challenge data description
Sepsis definition	Sepsis-3 (SOFA $\geq$ 2-point change)	Singer et al., 2016
Prediction horizon	6 hours before clinical recognition	Challenge data description
Teams / algorithms submitted	104 teams / 853 algorithms	Bennett et al., 2021
Best published Utility Score (final)	0.364 (team 'SailOcean')	Bennett et al., 2021

On reproducibility: the figures above are the officially published statistics of the dataset and challenge, not results produced by this project. The project's own pipeline (`src/preprocessing.py`) performs a stratified 70/15/15 train/validation/test split and automatically writes an exact dataset report (`data/processed/dataset_report.md`) once run against the downloaded data — see `docs/DATASET_CARD.md` for full detail on why no fabricated numbers are presented in this repository.

4. Methodology

4.1 Preprocessing

- Forward-fill imputation per patient, followed by cohort-median imputation for any remaining gaps (standard practice for this dataset).
- Z-score normalization of all 34 clinical features.
- Sliding 12-hour windows per patient-hour, labeled positive if sepsis onset occurs within the next 6 hours (the challenge's early-prediction criterion).
- Stratified 70/15/15 train/validation/test split preserving class balance.

4.2 Branch 1 — Temporal model (BiGRU + Attention)

A 2-layer bidirectional GRU (hidden size 128) processes the 12x34 input window. A learned attention layer produces a weighted context vector over the 12 hourly hidden states, which both feeds the classifier head and provides a natural per-hour importance score exposed in the API response (`top_contributing_signals`).

4.3 Branch 2 — Computer vision (GAF/Recurrence-Plot + CNN)

The same window is transformed into a 2-channel image: a Gramian Angular Field and a recurrence plot, computed for the four most clinically salient vitals and averaged into two channels. A small ResNet-style CNN (stem + 2 residual blocks + global average pooling) produces a 128-d embedding from this image.

4.4 Fusion

The GRU context vector and CNN embedding are concatenated and passed through a small MLP head (LayerNorm -> 128 -> 32 -> 1) to produce the final sepsis probability. Class imbalance is handled via a weighted random sampler and focal loss ($\alpha=0.85$, $\gamma=2.0$) during training.

4.5 System architecture

The project strictly separates the AI system from the web client: all model code, weights, and inference logic live behind a FastAPI service (`src/serve_api.py`); the static website (`webapp/`) only ever calls documented REST endpoints (`/predict`, `/explain`, `/admin/stats`). No model weights or inference code are ever shipped to the browser.

5. Explainable AI

Two complementary explanation mechanisms are implemented in `src/explain.py`:

- **SHAP-based (`explain_with_shap`)**: a GradientExplainer applied to the GRU branch's input, using a background sample from the training set. Produces per-feature importance scores with the theoretical consistency guarantees of Shapley values. Requires a trained model checkpoint.
- **Delta-based fallback (`explain_with_deltas`)**: a lightweight, dependency-free heuristic that compares the first and last quarter of the input window for each clinically salient variable (HR, MAP, Resp, Temp), producing a ranked, weighted list of contributing factors even when no trained model is available (e.g., in the live demo).

Both mechanisms feed the same API contract (`POST /explain`) and the same frontend rendering (a horizontal weighted bar chart on the patient page), so switching from the demo heuristic to real SHAP explanations once a model is trained requires no frontend changes.

6. System Deployment and Monitoring

6.1 Containerization

Two Dockerfiles are provided: `Dockerfile.api` (FastAPI + PyTorch inference service) and `Dockerfile.web` (nginx serving the static site). `docker-compose.yml` orchestrates both plus an optional monitoring profile (Prometheus + Grafana).

6.2 Cloud deployment targets

Ready-made configuration is provided for Render (`render.yaml`) and Railway (`railway.toml`), with a written guide (`DEPLOYMENT.md`) covering AWS (ECS Fargate / App Runner + S3/CloudFront) and Azure (Container Apps + Static Web Apps). No project resources have been deployed to a live public URL by the authors at the time of writing; this is stated explicitly rather than implied.

6.3 Monitoring and logging

Every prediction served by the API is logged to a local SQLite table (patient id, probability, risk level, timestamp). An admin endpoint (GET `/admin/stats`) aggregates total predictions, average risk, critical-case count, and an hourly breakdown, rendered in the `webapp/admin.html` dashboard. A `/metrics` endpoint exposes Prometheus-format counters (`sepsiswatch_predictions_total`, `sepsiswatch_high_risk_total`, prediction latency histogram) for production-grade observability via Prometheus/Grafana.

7. Clinical Validation — An Honest Assessment

The single most important question for any clinical AI project is: how do we know this is actually medically useful? At the time of writing, **no clinical validation has been performed** on this project — this is stated plainly rather than glossed over.

7.1 What has not been done

- No model has been trained on the real dataset in this environment (no network access to physionet.org, no GPU training run).
- No prospective (silent-mode) deployment against real patient outcomes.
- No clinician review of model outputs.
- No IRB/ethics committee approval, and no regulatory (e.g., FDA SaMD) submission.

7.2 The validation methodology that would need to be followed

- **Quantitative:** train on the provided pipeline, evaluate on the held-out test split, compute the official challenge Utility Score, and compare directly against the published leaderboard (best published score: 0.364, arXiv:2012.11013) as an external, objective benchmark.
- **Qualitative:** structured clinician review of a sample of predictions using the provided feedback template (docs/DOCTOR_FEEDBACK_TEMPLATE.md), covering timing usefulness, agreement with clinical judgment, and alarm-fatigue risk.
- **Prospective:** silent-mode deployment alongside standard care (predictions logged but not acted upon) over a period of months, followed by IRB approval and regulatory review before any predictions are allowed to influence care decisions.

Full detail is maintained in docs/CLINICAL_VALIDATION.md, which is treated as a living document to be updated as each validation stage is actually completed — not aspirational language, but a checklist against which real progress can be tracked.

8. Limitations

- **No trained model in this repository.** Code is complete and tested for syntax/logic, but weights (models/*.pt) must be produced by the user running the training pipeline on real data.
- **Missingness handling is simplistic.** Forward-fill + median imputation is a reasonable baseline but does not model measurement-uncertainty or missingness-as-signal explicitly.
- **Single-institution generalization risk.** As the official challenge showed (score dropping to -0.048 on the unseen hospital C), models trained on hospitals A/B may not generalize to a new hospital's practice patterns without careful domain adaptation.
- **The 'AI insights' shown on the demo dashboard are rule-based, not model-derived,** and exist purely to illustrate what a decision-support output layer could look like; they are explicitly labeled as such in the interface and are not a substitute for the trained model's actual output.
- **No live deployment exists at present** — deployment configuration is provided and untested against a real cloud account by the authors.

9. Future Work

9.1 Real-time ICU streaming

Move from hourly batch-style windows to a streaming architecture (e.g., Kafka or MQTT ingestion from bedside monitors) with incremental model updates, reducing the effective latency between a physiological change and a risk-score update from ~1 hour to near-real-time.

9.2 Integration with Hospital Information Systems (HIS/EHR)

Direct HL7 FHIR integration would remove manual data entry, allow the model to consume the full breadth of the electronic health record (medication administration, nursing notes via NLP, prior history), and allow risk scores to be written back into the clinician's existing workflow rather than a separate dashboard.

9.3 Federated learning across hospitals

Given the generalization gap observed in the original challenge (hospital C), federated learning would allow the model to learn from multiple institutions' data without that data ever leaving each hospital's infrastructure — directly addressing both the generalization problem and patient privacy/data-governance constraints.

9.4 Transformer-based architectures

Replacing or augmenting the BiGRU branch with a time-series transformer (e.g., a continuous-time or irregularly-sampled-aware attention architecture) may better capture long-range dependencies and irregular sampling patterns than a purely recurrent model, at the cost of requiring substantially more training data and compute.

9.5 Multi-hospital validation

Systematic external validation on hospital systems not used in training — ideally several, spanning different EHR vendors, patient populations, and care protocols — is the single highest-value next step for establishing whether this approach generalizes beyond the two public training hospitals.

10. Conclusion

SepsisWatch demonstrates a complete, technically sound pipeline for early sepsis prediction that combines temporal deep learning with a genuine computer-vision branch, wraps it in an explainable, monitored, deployable system, and — most importantly — is explicit about what has and has not been established. The project's value at this stage is as a rigorous educational reference architecture and a concrete starting point for a real validation effort, not as a clinically validated product. The path from here to clinical usefulness is well-defined (Section 7) and the codebase is structured so that each step — training on real data, quantitative benchmarking, clinician review, prospective silent deployment — can be executed without architectural rework.

References

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